Manufacturing Use Case

Document Information

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Executive Summary

This case study illustrates how a representative discrete manufacturing company implemented data governance to improve supply chain visibility, product data accuracy, and operational reporting. It is intended as an illustrative reference — adapt the specifics to your own organization's context.

Organization Profile: Mid-size industrial manufacturer, multiple plants, global supplier network, ERP plus plant-level MES systems.

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1. Business Context

1.1 The Challenge

- Product and materials master data was inconsistent across ERP, MES, and supplier portals, causing procurement and inventory discrepancies
- No single source of truth for bill-of-materials (BOM) data led to production errors and rework
- Supply chain visibility was limited by inconsistent supplier and part identifiers across systems
- Plant-level reporting could not be reliably aggregated for enterprise-wide operational dashboards

1.2 Business Drivers

- Cost pressure from inventory inaccuracy and production rework
- Customer and regulatory requirements for supply chain traceability (e.g., country-of-origin, materials compliance)
- Enterprise digital transformation initiative requiring trusted data for analytics and automation

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2. Governance Approach

2.1 Program Structure

The Data Governance Council was sponsored by the COO and VP of Supply Chain, reflecting the program's operational rather than purely IT focus.

2.2 Phased Rollout

1. **Foundation (Months 1-3):** Charter approval; classification scheme tailored to product, materials, and supplier data
2. **Pilot (Months 4-7):** Materials Master Data domain selected as pilot across two plants
3. **Expansion (Months 8-14):** Extended to BOM data and supplier master data across all plants
4. **Operationalization (Months 15-18):** Governance metrics embedded into operations and supply chain reviews

2.3 Key Capabilities Implemented

- Unified materials master data catalog spanning ERP and plant-level MES systems
- Data quality rules for part number standardization and duplicate detection
- Lineage tracking from supplier data feeds through to BOM and production reporting
- Plant-level Technical Data Stewards paired with a central Business Data Steward for materials data

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3. Results

Metric	Before	After (18 Months)
Duplicate part number rate	~9%	<2%
Inventory discrepancy incidents (monthly)	45	12
Time to assemble enterprise-wide production dashboard	2 weeks	2 days
Plants with standardized materials master data	0 of 6	6 of 6

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4. What Worked Well

- Operational (not just IT) sponsorship gave the program credibility on the plant floor
- Starting with materials master data addressed a pain point felt directly by procurement and production planning
- Standardizing part number formats early simplified every downstream governance effort

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5. Challenges Encountered

- Plant-level MES systems varied significantly in technical maturity, requiring different integration approaches per plant

- Supplier-provided data quality varied widely, requiring a supplier data quality scorecard to be added mid-program
- Initial materials classification scheme had to be revised after pilot feedback from plant operations teams

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6. Lessons for Other Manufacturing Organizations

- Materials and product master data is often the highest-value starting domain — prioritize it for early wins
- Expect significant variation in plant-level system maturity; build flexibility into the technical integration approach
- Extend data quality expectations to suppliers, not just internal systems, especially for traceability-sensitive industries

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Document Control:

- This is an illustrative template case study for adaptation to your organization
- Replace illustrative metrics and details with your own organization's data before external use
- Pair with Vendor Selection Criteria when evaluating supplier data integration tooling